

Module 1: Introduction to Black Holes

LATEST SUBMISSION GRADE

100%

1. What is a black hole?

1 / 1 point

- ☐ A large mass made out of dark matter.
- ☐ An optical illusion caused when a star absorbs light.
- ☐ A gateway to distant parts of the universe.
- ☒ The remnant of a dead star that has undergone a gravitational collapse.



Correct

Yes! A black hole forms when a star dies and its matter is compressed by gravity.

2. Which movie or TV show attempts to portray the most scientifically accurate version of a black hole?

1 / 1 point

- ☒ Interstellar (2014)
- ☐ Star Trek - The Future Begins (2009)
- ☐ The Black Hole (1979)
- ☐ Doctor Who - The Impossible Planet (2006)



Correct

Yes! Christopher Nolan worked closely with Kip Thorne in order to produce the most scientifically accurate portrayal of a black hole.

3. Which word describes the fear of black holes?

1 / 1 point

- ☐ Deinophobia
- ☒ Melanoheliophobia
- ☐ Orophobia
- ☐ Entomophobia



Correct

Yes! Melanoheliophobia is the fear of black holes!

4. What size would a black hole be if it was the same mass as Earth?

1 / 1 point

- ☐ The size of a large city.
- ☐ As small as an atom.
- ☐ The same size as the Earth.
- ☒ About the size of a marble.

✓ **Correct**

Yes! The mass of the Earth must be compressed smaller than 1 cm to form a black hole.

5. Who was the first person to describe the properties of a black hole?

1 / 1 point

- ☒ John Mitchell (1783)
- ☐ Stephen Hawking (1970)
- ☐ Isaac Newton (1679)
- ☐ Galileo Galilei (1589)

✓ **Correct**

Yes! John Mitchell first described black holes by setting escape velocity to the speed of light.

6. A black hole consists of a central _____, with a boundary called the _____.

1 / 1 point

- ☐ point, sphere of no return
- ☐ star, escape velocity surface
- ☐ dark sphere, black hole sphere
- ☒ singularity, event horizon

✓ **Correct**

Yes! The singularity is at the centre, and the event horizon limits what we can see of it.

7. The correct value for G, Newton's Gravitational Constant, is

1 / 1 point

- ☐ 11.2 km/s
- ☐ 9.81 m/s²
- ☒ $6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2$
- ☐ $6.626 \times 10^{-34} \text{ J} \cdot \text{s}$

✓ **Correct**

Yes! This is the value of Newton's Gravitational Constant.

8. Consider the equation for Newton's Law of Universal Gravitation. If mass M is **doubled**, the force between the two objects will _____. If the distance between the objects is **doubled**, the force between the objects will _____.

$$F = \frac{GMm}{r^2}$$

- ☐ increase by a factor of 2; increase by a factor of 4
- ☐ decrease by a factor of 2; increase by a factor of 4
- ☐ decrease by a factor of 2; decrease by a factor of 4
- ☒ increase by a factor of 2; decrease by a factor of 4

✓ **Correct**

Yes! Doubling the mass doubles the force, doubling the distance quarters the force.

9. The maximum achievable speed in the universe is 1 / 1 point

- ☒ the speed of light, as measured in a vacuum
- ☐ the speed of light, as measured in Earth's atmosphere.
- ☐ the speed of sound, as measured in vacuum.
- ☐ the speed of sound, as measured deep inside the Sun.

✓ **Correct**

10. A "Dark Star" is a star whose escape velocity has exceeded 1 / 1 point

- ☐ the speed of darkness, at the star's surface.
- ☐ the orbital speed of the galactic core.
- ☐ the speed of light, in the centre of the star.
- ☒ the speed of light, at the star's surface.

✓ **Correct**

Yes! Light cannot escape from a dark star surface.